DIANA Clustering Algorithm — Easy Numerical Example

DIANA is a **divisive hierarchical clustering** method:

- Start with **all data in one big cluster**.
- Then **repeatedly split** the cluster by finding the object **most dissimilar** from the rest.
- That object becomes the **seed of a new cluster**.
- Continue moving objects based on dissimilarity.

Dataset

We use a very small dataset of **4 objects**, and a simple **distance matrix**.

Objects	A	B	C	D
A	0	3	10	12
B	3	0	8	9
C	10	8	0	2
D	12	9	2	0

Interpretation:

- Smaller number → objects are similar.
- Larger number → objects are far apart.

STEP 1 — Start with one cluster

C_0 = {A, B, C, D}

STEP 2 — Find the object farthest from the others

We compute **average dissimilarity** of each object with all others:

For A

(3 + 10 + 12) / 3 = 25/3 = **8.33**

For B

(3 + 8 + 9) / 3 = 20/3 = **6.67**

For C

(10 + 8 + 2) / 3 = 20/3 = **6.67**

For D

(12 + 9 + 2) / 3 = 23/3 = **7.67**

Farthest object = A (8.33)

So **A starts the first splinter cluster**.

Splinter cluster (S):

S = {A}

Remaining cluster (R):

R = {B, C, D}

STEP 3 — Decide which object joins splinter cluster

For each object in R (B, C, D):

We calculate:

d(obj, S) – d(obj, R)

If the value is **positive**, the object is **more dissimilar to R → should join S**.

For B

d(B, A) = 3
Average d(B, R) = avg(8, 9) = 8.5
Difference = 3 − 8.5 = **−5.5**
→ stays in R

For C

d(C, A) = 10
Average d(C, R) = avg(8, 2) = 5
Difference = 10 − 5 = +**5**
→ joins S

For D

d(D, A) = 12
Average d(D, R) = avg(9, 2) = 5.5
Difference = 12 − 5.5 = +**6.5**
→ joins S (largest positive)

Updated Splits

Splinter cluster:

S = {A, C, D}

Remaining cluster:

R = {B}

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STEP 4 — Stopping rule

Once R has only **one element left**, we **stop splitting this level**.

So the first split is:

C_0 = {A, B, C, D} → {B} and {A, C, D}

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STEP 5 — Continue splitting the larger cluster {A, C, D}

We now apply DIANA again on **{A, C, D}**.

We compute average dissimilarity:

For A: (10 + 12)/2 = 11

For C: (10 + 2)/2 = 6

For D: (12 + 2)/2 = 7

Farthest object = **A** → A becomes splinter seed.

S = {A}

R = {C, D}

Which joins S?

For C

d(C, A) = 10
avg d(C, R) = 2
10 − 2 = +**8** → joins S

For D

d(D, A) = 12
avg d(D, R) = 2
12 − 2 = +**10** → joins S (larger)

So S = {A, C, D} again.

Since R becomes empty, we end.

So final split is:

{A, C, D} → {A} and {C, D}

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FINAL DIANA DENDROGRAM CLUSTERS

The splits are:

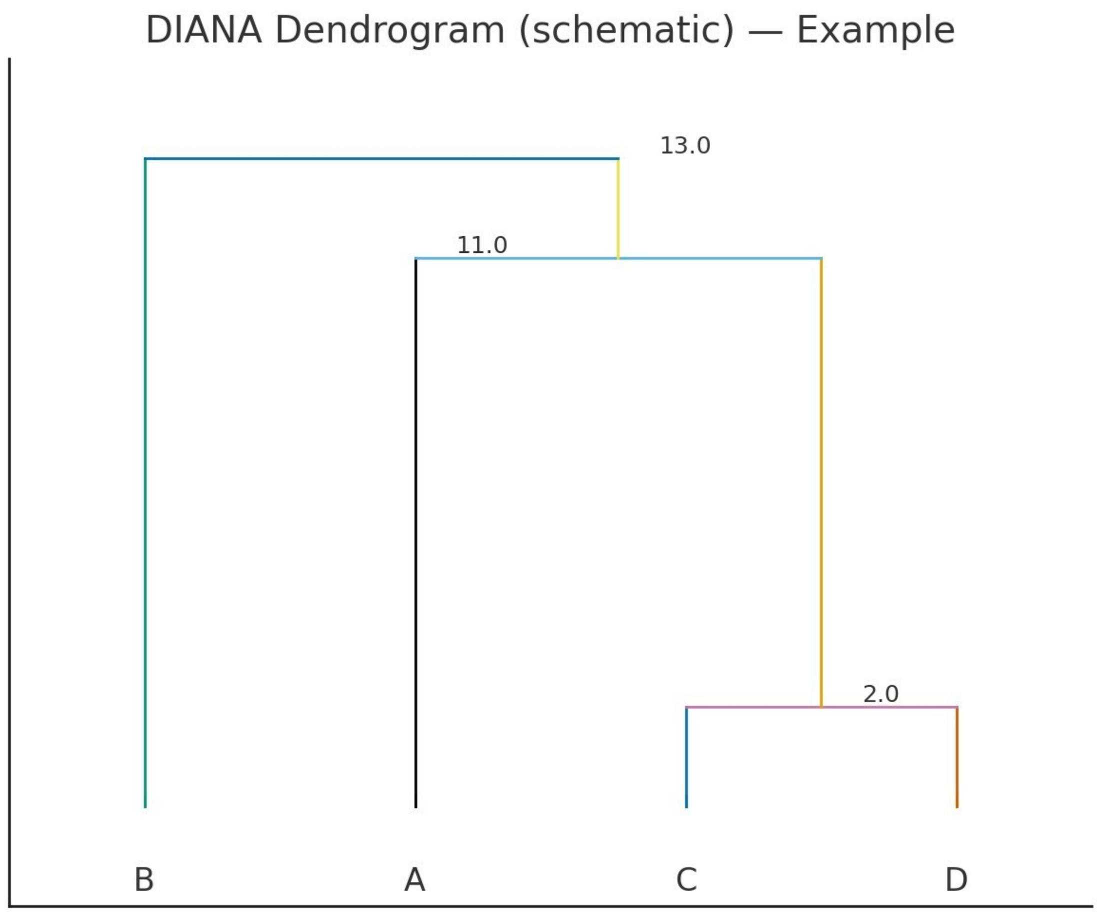
1. **{A, B, C, D}**
   → {B} and {A, C, D}
2. **{A, C, D}**
   → {A} and {C, D}

Final Clustering Output

- **Cluster 1:** {B}
- **Cluster 2:** {A}
- **Cluster 3:** {C, D}

Because:

- C and D are most similar (distance = 2)
- B is closest to A but still separated early
- A is very dissimilar from others



a schematic DIANA dendrogram for the example and displayed it above.

Quick note: the plot is a visual, not a numeric linkage output from a library. The branch heights (labels) are chosen to reflect the split order and relative dissimilarities from the numerical example:

- C and D merge at height 2 (distance = 2).
- A separates from {C, D} at height ≈ 11 (average of distances 10 and 12).
- B splits from {A, C, D} at the top.

- A dendrogram with exact numeric heights computed using a particular linkage rule (single/complete/average).
- An animated step-by-step build of the DIANA splits.

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